

Lightweight Test-Time Q Calibration for Frozen Meta-RL under Dynamics Shift

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Abstract

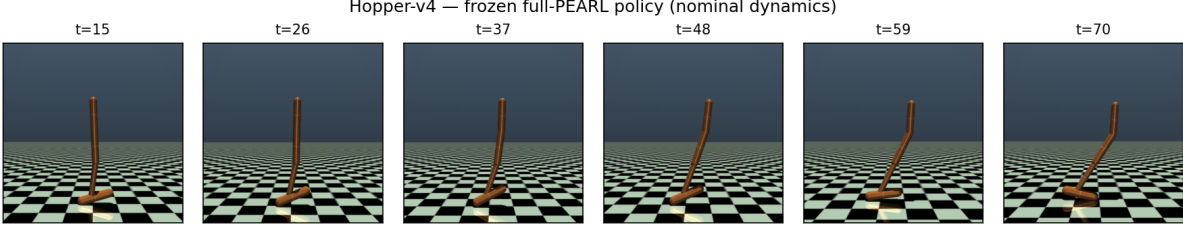
Meta-reinforcement learning methods such as PEARL adapt at test time by inferring a latent task variable from recent context. Under out-of-distribution dynamics shift, however, latent adaptation alone may be insufficient: the frozen policy may still propose useful local actions, while the frozen critic misranks them. This report studies a lightweight alternative to policy or critic fine-tuning. We freeze the PEARL encoder, actor, and critic backbone, and update only a small Bayesian linear head over the critic’s penultimate features. The head is trained online to predict a Bellman-residual correction and is used to re-rank candidate actions proposed near the frozen policy. Across Hopper-v4 dynamics shifts, the method improves moderate action-effectiveness, friction, compound, and non-stationary regimes with small nominal cost. Against test-time fine-tuning, its strongest advantage is not asymptotic return in every regime; rather, it is deployment safety: a single precommitted configuration has much smaller worst-case regret, and in zero-warmup online evaluation it achieves the highest cumulative return on all tested OOD regimes from the first few episodes.

1 Motivation

The central question is whether a frozen meta-RL policy can adapt to test-time dynamics shift without policy fine-tuning or full model-based planning. We focus on a PEARL-style latent-conditioned SAC backbone trained on Hopper-v4 dynamics randomization. At deployment, the policy and critic are frozen. The environment may have shifted action effectiveness, contact/friction, body mass, or non-stationary dynamics.

The working hypothesis is that many moderate OOD shifts are not failures of the policy proposal distribution itself. The frozen policy may still generate recoverable local actions, but the frozen critic may assign them the wrong relative values. If this is true, a cheap correction to the critic ranking can provide adaptation without changing the actor.

The method, denoted as **QC-PEARL**, uses the Bellman residual of the frozen critic as a test-time signal. The correction is intentionally small: it is a linear Bayesian recursive estimator on learned critic features, combined with a shift gate that limits deviation from the base policy when the environment looks nominal.



2 Frozen PEARL Backbone

Let $s_t \in \mathbb{R}^{d_s}$ and $a_t \in \mathbb{R}^{d_a}$ denote the state and action. For Hopper-v4, $d_s = 11$ and $d_a = 3$. PEARL represents the task through a latent variable $z \in \mathbb{R}^{d_z}$, with $d_z = 5$ in the experiments. A product-of-Gaussians context encoder maps recent context c_t to a posterior over z ,

$$z_t \sim q_\psi(z \mid c_t), \quad (1)$$

and a latent-conditioned tanh-Gaussian actor proposes actions,

$$a_t \sim \pi_\theta(a \mid s_t, z_t), \quad a_0(s_t, z_t) = \mu_\theta(s_t, z_t). \quad (2)$$

The critic consists of twin Q-functions,

$$Q_{\min}(s, a, z) = \min_{i \in \{1, 2\}} Q_{\phi_i}(s, a, z). \quad (3)$$

The backbone is meta-trained on a dynamics-randomized task distribution

$$\xi = (\xi_{\text{mass}}, \xi_{\text{fric}}, \xi_{\text{damp}}, \xi_{\text{act}}), \quad \xi_j \sim \mathcal{U}([0.8, 1.2]), \quad (4)$$

where the components scale body mass, friction, damping, and action effectiveness. At test time, q_ψ , π_θ , Q_{ϕ_1} , and Q_{ϕ_2} remain frozen. The offline world model $f_\eta(s, a, z)$ is also frozen. Only the online residual estimators are updated.

3 Bellman Inconsistency under Dynamics Shift

Define the deterministic bootstrap value

$$V(s, z) = Q_{\min}(s, \mu_\theta(s, z), z). \quad (5)$$

For an executed transition $(s_t, a_t, r_t, s_{t+1}, d_t)$, the frozen-critic Bellman residual is

$$\eta_t = r_t + \gamma(1 - d_t)V(s_{t+1}, z_t) - Q_{\min}(s_t, a_t, z_t). \quad (6)$$

Under the training distribution, the critic is trained to be approximately Bellman-consistent, so $\eta_t \approx 0$ in familiar regions. Under a dynamics shift, $p_{\text{test}}(s' \mid s, a, z)$ differs from the training distribution, the inferred latent z_t may be biased or insufficient, and Q, V may extrapolate poorly. Thus η_t becomes a direct local indicator of critic inconsistency.

This residual has two roles. First, a cheaper dynamics-model residual is used to gate how aggressively the method deviates from the frozen policy. Second, a value-residual target derived from (6) is used to fit the online Q-calibration head.

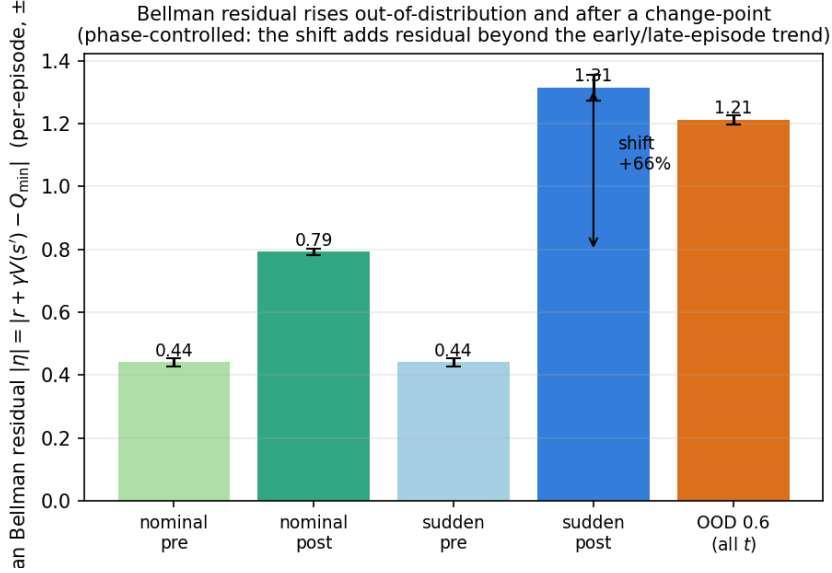


Figure 1: Frozen-critic Bellman residual under nominal, fixed OOD, and sudden-shift regimes. The residual increases under OOD dynamics and rises after the sudden change point, indicating that Bellman inconsistency provides a useful test-time signal for detecting and correcting value miscalibration.

Figure 1 shows that the frozen critic’s Bellman residual is informative about test-time distribution shift. In the nominal regime, the residual remains relatively small, indicating that the frozen critic is approximately Bellman-consistent under the training dynamics. Under fixed OOD dynamics, the residual increases substantially, suggesting that the frozen value function is no longer self-consistent with the realized transition and reward. In the sudden-shift regime, the residual is low before the change point and rises clearly after the actuator change, which supports using the residual as both a shift detector and a calibration target.

A useful caveat is that the residual is not a pure change-point statistic by itself. Even in the nominal regime, later parts of a long episode can naturally produce larger residuals because the agent visits harder or less frequently sampled states. Therefore, for non-stationary regimes, the relevant comparison is not only “post-shift versus pre-shift” within the same episode, but also “post-shift versus nominal at the same episode phase.” After controlling for this long-horizon phase effect, the sudden regime still shows a clear residual increase after the dynamics change. Thus, the residual contains a genuine shift-induced signal, while the gate should interpret it with temporal smoothing rather than as a raw instantaneous alarm.

4 Method: Online Bellman-Residual Q Calibration

4.1 Candidate Action Pool

At each state s_t , the frozen policy produces a base deterministic action

$$a_0 = \mu_\theta(s_t, z_t). \quad (7)$$

We build a local candidate pool $\mathcal{A}_t = \{a_i\}_{i=1}^K$ consisting of a_0 , Gaussian perturbations around a_0 , and samples from $\pi_\theta(\cdot | s_t, z_t)$, clipped to the action bounds. The method does not search the full action space; it only re-ranks plausible local actions proposed near the frozen policy.

4.2 Residual Decomposition

Let $f_\eta(s, a, z)$ be the frozen offline dynamics model, predicting the nominal state increment. The nominal model next state is

$$\hat{s}'_{\text{nom}} = s + f_\eta(s, a, z). \quad (8)$$

A full one-step shifted value estimate could be written as

$$\hat{\eta}(s, a) = \hat{r}(s, a, \hat{s}') + \gamma V(\hat{s}', z) - Q_{\min}(s, a, z), \quad (9)$$

so every calibrated score can be interpreted as $Q_{\min}$ plus an estimate of a Bellman residual.

The implementation separates the residual into a dynamics-driven part and an intrinsic value-error part:

$$\hat{\eta}(s, a) = \underbrace{\hat{r}(s, a, \hat{s}'_{\text{nom}}) + \gamma V(\hat{s}'_{\text{nom}}, z) - Q_{\min}(s, a, z)}_{\eta_{\text{value}}(s, a)} + \underbrace{\Delta r + \gamma \Delta V}_{\text{dynamics value shift}}. \quad (10)$$

The QC-PEARL method focuses on the intrinsic value-error term η_{value} , because this is the part that directly corrects critic misranking in the learned feature basis.

4.3 Q-Feature Residual Head

Let $\varphi^Q(s, a, z)$ denote the penultimate feature vector of the frozen critic. Since the critic's last layer is linear in φ^Q , a natural local approximation to the value error is

$$h_w(\varphi^Q(s, a, z)) = w^\top \varphi^Q(s, a, z). \quad (11)$$

The head is fit online from executed transitions. The target used in the implementation is

$$y_t = r_t + \gamma(1 - d_t)V(s_t + f_\eta(s_t, a_t, z_t), z_t) - Q_{\min}(s_t, a_t, z_t). \quad (12)$$

This uses the real observed reward but a nominal-model next state in the bootstrap. The purpose is to estimate the value-error component without double-counting the explicit dynamics shift.

For a candidate action a_i , the calibrated score is

$$\text{score}_i = Q_{\min}(s_t, a_i, z_t) + h_w(\varphi^Q(s_t, a_i, z_t)) - \Omega(a_i), \quad (13)$$

where the local trust-region penalty is

$$\Omega(a_i) = \lambda_a \|a_i - a_0\|^2 + \lambda_{\text{sw}} \|a_i - a_{t-1}\|^2. \quad (14)$$

The selected candidate is

$$a_{\text{sel}} = \arg \max_{a_i \in \mathcal{A}_t} \text{score}_i. \quad (15)$$

4.4 Shift Gate

To avoid unnecessary correction in nominal regimes, the executed action interpolates between the frozen policy and the calibrated candidate:

$$a_t = (1 - h_t)a_0 + h_t a_{\text{sel}}. \quad (16)$$

The gate is based on the one-step dynamics-model residual

$$\varepsilon_t = \|(s_{t+1} - s_t) - f_\eta(s_t, a_t, z_t)\|. \quad (17)$$

Let ε_{p95} be the nominal 95th percentile residual scale. The normalized EWMA and gate are

$$\bar{\varepsilon}_t = (1 - \rho_g)\bar{\varepsilon}_{t-1} + \rho_g \frac{\varepsilon_t}{\varepsilon_{p95}}, \quad h_t = \sigma(\kappa(\bar{\varepsilon}_t - \tau_g)). \quad (18)$$

Thus the method reverts toward the frozen policy when the observed transition stream looks close to nominal.

4.5 Bayesian Recursive Linear Update

The online residual head is a Bayesian recursive linear regressor with an AR(1) prior. Let M be the coefficient mean and P the posterior covariance. For feature $\phi_t = \varphi^Q(s_t, a_t, z_t)$ and target y_t , the prediction step is

$$M \leftarrow \rho_q M, \quad P \leftarrow \rho_q^2 P + q_\alpha I. \quad (19)$$

The measurement update is

$$S_t = \sigma_y^2 + \phi_t^\top P \phi_t, \quad (20)$$

$$K_t = P \phi_t / S_t, \quad (21)$$

$$M \leftarrow M + K_t (y_t - M^\top \phi_t), \quad (22)$$

$$P \leftarrow P - K_t \phi_t^\top P. \quad (23)$$

The predictive mean and variance for a candidate feature ϕ are

$$\hat{y}(\phi) = M^\top \phi, \quad \text{Var}[\hat{y}(\phi)] = \phi^\top P \phi. \quad (24)$$

This posterior variance also gives an optional lower-confidence-bound score,

$$\text{score}_i^{\text{LCB}} = Q_{\min}(s_t, a_i, z_t) + h_w(\varphi_i^Q) - \beta \sqrt{\varphi_i^{Q^\top} P \varphi_i^Q - \Omega(a_i)}, \quad (25)$$

which acts as a risk-averse correction when the residual estimate is uncertain.

5 Baselines and Evaluation Protocols

Frozen PEARL. FULL-PEARL executes $a_0 = \mu_\theta(s, z)$ with PEARL-style latent context adaptation but no Q re-ranking.

Q-Greedy. Q-GREEDY re-ranks the same candidate pool by the raw frozen critic,

$$\text{score}_i = Q_{\min}(s_t, a_i, z_t) - \Omega(a_i), \quad (26)$$

and therefore isolates how much improvement comes from candidate re-selection alone.

Value-shift QR / QC-PEARL. QC-PEARL uses (13) with the Bayesian Bellman-residual head. The encoder, actor, critic backbone, and world model remain frozen.

Test-time fine-tuning. Two fine-tuning baselines are used. FT-LASTLAYER updates only the last linear layer of the critic by SGD and acts by candidate re-ranking with the fine-tuned critic. FT-FULL updates actor and critic weights by SAC-style fine-tuning, with the encoder frozen. Both use the same in-regime adaptation data budget as QC-PEARL and never overwrite the original pretrained checkpoint.

Table 1: Main OOD return comparison. Results are averaged over three seeds. The columns “QR-PEARL” and “QR-QG” report relative improvement of QC-PEARL over frozen PEARL and Q-GREEDY.

Regime	PEARL	Q-greedy	Ours	QR-PEARL	QR-QG
nominal	1468	1452	1447	−1.5%	−0.4%
action gain 0.60	721	788	842	+16.8%	+6.8%
action gain 0.40	371	371	362	−2.3%	−2.4%
action bias 0.20	978	1116	1085	+11.0%	−2.7%
action noise 0.20	1485	1459	1450	−2.4%	−0.6%
friction 0.60	864	874	911	+5.4%	+4.2%
friction 1.50	424	417	432	+1.9%	+3.5%
mass 0.60	1426	1280	1389	−2.6%	+8.6%
compound mild	915	992	1048	+14.5%	+5.7%
compound hard	515	551	552	+7.2%	+0.1%
sudden 1.0 → 0.4 @ $t = 100$	604	607	614	+1.7%	+1.2%
gradual 1.0 → 0.2	689	710	727	+5.5%	+2.3%
multi-sudden	648	642	726	+12.1%	+13.1%

OOD regimes. The test regimes include nominal dynamics, action gain shifts, action bias/noise, friction shifts, mass shifts, compound shifts, and non-stationary schedules such as sudden, gradual, and multi-sudden actuator changes. The main OOD table uses three seeds with fleet breadth $M = 5$. The fine-tuning robustness comparison uses $K = 3$ warmup episodes, $M = 3$ fleet data parity, and three seeds. The online cumulative-return comparison uses $K = 0$, $M = 1$, and scores every deployment episode from episode 1.

6 Results

6.1 Main OOD Results

Table 1 reports the main return comparison over OOD type. QC-PEARL is strongest on regimes where the frozen policy still proposes useful local actions but the critic ranking is biased: moderate action gain, friction, compound shifts, and some non-stationary schedules. It is not a universal fix. Severe actuator loss remains difficult, and pure action bias is better handled by raw Q-greedy candidate selection.

The nominal cost of QC-PEARL is small, about 1–2% in this table. The gain is largest when the shift changes the critic’s value calibration without making the frozen action proposal distribution useless. Under strong actuator degradation, the candidate pool itself may not contain recoverable actions; in that case a Q-residual re-ranker cannot repair the policy-level failure.

6.2 Fine-Tuning Comparison: Robustness under No Validation Set

A fair test-time-adaptation protocol cannot tune hyperparameters on the unknown deployment shift. The method must commit one configuration before seeing which OOD regime occurs. Therefore, the key metric is the worst-case regret relative to frozen PEARL across regimes:

$$\text{WorstRegret}(m) = \min_{r \in \mathcal{R}} [J_m(r) - J_{\text{PEARL}}(r)]. \quad (27)$$

Here $\mathcal{R} = \{\text{nominal, action gain 0.6, sudden} \rightarrow 0.4, \text{multi-sudden}\}$. Table 2 shows that QC-PEARL has near-zero worst-case regret and is insensitive to its prior-variance knob. Fine-tuning can obtain larger gains on the repeated-change multi-sudden regime, but every fine-tuning configuration pays a much larger nominal safety tax.

Table 2: Fine-tuning comparison under a no-validation-set protocol. Entries are regret versus frozen PEARL return. WORST is the minimum across regimes for a single committed configuration. $K = 3$, $M = 3$, three seeds.

Config	nominal	ood 0.6	sudden 0.4	multi	WORST
QC-PEARL prior-var 0.10	-18	+117	+7	+35	-18
QC-PEARL prior-var 0.25	-16	+109	+7	+34	-16
QC-PEARL prior-var 1.00	-18	+99	+7	+61	-18
ft-lastlayer lr 10^{-4} u50	-119	+85	+0	+106	-119
ft-lastlayer lr 10^{-4} u200	-102	+72	+7	+127	-102
ft-lastlayer lr $3 \cdot 10^{-4}$ u50	-133	+81	-1	+107	-133
ft-lastlayer lr $3 \cdot 10^{-4}$ u200	-109	+65	-1	+131	-109
ft-full lr 10^{-4} u50	-189	+62	+27	+73	-189
ft-full lr 10^{-4} u200	-213	+70	-6	+106	-213
ft-full lr $3 \cdot 10^{-4}$ u50	-218	+48	-102	+81	-218
ft-full lr $3 \cdot 10^{-4}$ u200	-482	+48	-260	+54	-482

The main conclusion from [table 2](#) is not that QC-PEARL has the highest OOD return in every column. The more defensible claim is that it is the safest deployable test-time adaptation rule among the compared methods. It is equal or better than fine-tuning on nominal, action-gain 0.6, and sudden $\rightarrow$ 0.4, while FT-LASTLAYER can be better on multi-sudden only by accepting a large nominal penalty. FT-FULL is substantially more unstable.

6.3 Online Anytime Deployment: Cumulative Return

Post-warmup evaluation can hide a deployment cost. In a true zero-shot deployment there is no free warmup episode; every episode counts from step 1. We therefore evaluate cumulative return over the first E deployment episodes with $K = 0$ and $M = 1$. This directly measures whether adaptation is useful before its warm-up cost has been amortized.

Table 3: Online cumulative return with no warmup. Every deployment episode is scored. $K = 0$, $M = 1$, three seeds, $E = 10$. Δ vs frozen @10 is $\sum_{e \leq 10} (R_e^m - R_e^{\text{FULL-PEARL}})$.

Regime	Method	ep1	cum-ret@5	cum-ret@10	Δ vs frozen @10
nominal	frozen	1447	7337	14674	0
	q-greedy	1442	7247	14502	-172
	QC-PEARL	1439	7252	14501	-173
	ft-lastlayer	1443	7202	14474	-200
	ft-full	1447	6539	11690	-2984
action gain 0.60	frozen	717	3600	7210	0
	q-greedy	786	3927	7857	+647
	QC-PEARL	819	4167	8296	+1086
	ft-lastlayer	810	3935	7846	+636
	ft-full	717	3280	5803	-1407
compound mild	frozen	886	4577	9140	0
	q-greedy	997	4973	9936	+796
	QC-PEARL	1051	5235	10428	+1288
	ft-lastlayer	979	5122	9876	+736
	ft-full	886	4293	8621	-519
multi-sudden	frozen	648	3241	6492	0
	q-greedy	642	3220	6484	-8
	QC-PEARL	694	3561	6922	+430
	ft-lastlayer	647	3342	6753	+261
	ft-full	648	3050	5721	-771

Table 3 changes the interpretation of the fine-tuning comparison. In post-warmup or asymptotic multi-sudden evaluation, FT-LASTLAYER can eventually outperform QC-PEARL. In the online cumulative metric, however, QC-PEARL is ahead on all three OOD regimes, including multi-sudden. The reason is that the recursive Bayesian head corrects within the first episode, while SGD fine-tuning needs multiple episodes before its benefit catches up. Full actor-critic fine-tuning is especially costly in the anytime setting: it is worse than frozen PEARL within ten episodes on all four tested regimes.

The nominal row also matters. Among the safe methods, nominal cumulative return is essentially tied: QC-PEARL, Q-GREEDY, and FT-LASTLAYER all pay about a 1% anytime tax, while FT-FULL collapses. Thus the OOD cumulative gains of QC-PEARL are not purchased by a large nominal deployment penalty.

6.4 Uncertainty as a Conservative Correction Knob

Because QC-PEARL maintains a posterior covariance, it can penalize uncertain residual corrections through the LCB score. In the available sweep, increasing β from 0 to 2 was neutral on nominal and sudden regimes, increased action-gain return from 828 to 840, and increased multi-sudden return from 683 to 705 while reducing multi-sudden seed standard deviation from 17 to 11. The effect is near the three-seed noise floor, so the appropriate claim is modest: posterior uncertainty provides a principled conservatism knob that fine-tuning lacks structurally.

7 Interpretation

The experiments support four conclusions.

First, Bellman-residual calibration is useful when the policy proposal distribution remains adequate but the critic ranking is biased. This explains the gains on moderate action-gain, friction, compound, and several non-stationary shifts.

Second, the method is limited by the local candidate pool. If the frozen policy no longer proposes recoverable actions, as in strong actuator loss, re-ranking cannot solve the policy-level failure. This is not a bug of the estimator; it is a consequence of keeping the actor frozen and avoiding global planning.

Third, the strongest comparison against fine-tuning is a deployment-safety comparison, not a universal-return comparison. Fine-tuning has a higher ceiling in some repeated-change settings, but it is tuning-sensitive, pays a nominal tax, and can destabilize the actor. QC-PEARL instead has near-zero worst-case regret under a single precommitted configuration.

Fourth, the cumulative-return result is important. If every deployment episode counts, QC-PEARL is more attractive than post-warmup tables alone suggest. It improves OOD cumulative return immediately, while fine-tuning’s eventual benefit can arrive too late to dominate the first several episodes.

8 Limitations and Next Steps

The current evidence is based on Hopper-v4. The method should be validated on additional locomotion tasks and non-locomotion continuous-control domains. The result also depends on the quality of the local action candidate pool. Wider action search, latent adaptation, or actor-level adaptation may be needed for severe OOD shifts where the frozen policy cannot propose stabilizing actions.

A second limitation is statistical power. Several comparisons use three seeds, and the uncertainty-LCB sweep is near the noise floor. The main qualitative conclusions are consistent across metrics, but final claims should be phrased conservatively unless additional seeds confirm them.

Finally, the fine-tuning comparison suggests a possible hybrid direction, but the current hybrid results are negative in the anytime window. Adding SGD on top of the recursive head reintroduces fine-tuning instability. Future hybrids should preserve the safety mechanisms of QC-PEARL: a frozen trusted base policy, residual correction with uncertainty, and a gate that can revert to the base action.

9 Conclusion

QC-PEARL is a lightweight test-time adaptation method for frozen meta-RL. It does not fine-tune the encoder, actor, or critic backbone. Instead, it fits a Bayesian linear Bellman-residual head over critic features and uses it to re-rank local action candidates. The method improves several moderate OOD dynamics shifts with small nominal cost. Its main advantage over test-time fine-tuning is safety under realistic deployment constraints: no per-regime validation tuning, low worst-case regret, explicit uncertainty, and high cumulative return from the start of online deployment.

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